Kyle Aaron Thompson

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Education

Auburn University — Auburn, Alabama
Ph.D. Mechanical Engineering — *Expected 2028*
M.S. Mechanical Engineering — *August 2025*
B.S. Mechanical Engineering, Magna Cum Laude — *August 2020*

Experience

Graduate Research Assistant – Auburn University — August 2020 – Current
GPS and Vehicle Dynamics Laboratory (GAVLAB) — *Auburn, Alabama*

- Conduct sponsored research in guidance, navigation, and control (GNC) of autonomous ground vehicles, focusing on guidance and control, with additional work on aerial platforms.
- Serve as controls, guidance, and coding lead for Autonomous Tiger Racing (ATR), Auburn's Indy Autonomous Challenge (IAC) team.
- Built the team's raceline optimizers and optimal online feedback controllers, and developed the majority of the current software stack.
- Developed a GPS-independent leader–follower convoy system using UWB ranging and vehicle models, achieving 1–5 m path-duplication accuracy at 10–50 m following distances.
- Designed an intelligent emergency stop system that tracks minimum-jerk trajectories with an LQR path-tracking controller, targeted to run on a low-cost backup processor.
- Published multiple academic papers as first author and co-authored additional conference and journal papers.
- Assisted in the laboratory instruction of undergraduate courses taught at Auburn University.

United States Marine Corps Reserves — July 2014 – Current
Gunnery Sergeant

- Oversee safe operation of M142 HIMARS rocket artillery systems for Battery K, 2nd Battalion, 14th Marines.
- Train and supervise an Operational Firing Platoon of approximately 35 Marines.
- Developed and taught curriculum for the professional certification of HIMARS operators.
- Deployed to Afghanistan with Task Force Southwest 19.1 and 19.2 (2019) and to Okinawa (2022–2023).

Teledyne Brown Engineering, Inc. — April 2015 – July 2015; June 2016 – August 2016
Classified Lab Administrative Assistant — *Huntsville, Alabama*

- Opened and secured the classified lab after hours and enforced government and company security policies.
- Performed basic system administration for classified lab users and supported software documentation.

Selected Publications

- **K. Thompson**, "Intelligent Emergency Stop System for Autonomous Ground Vehicles Using Minimum Jerk Trajectories and Path Tracking Linear Quadratic Regulator," *M.S. thesis*, Auburn University, Auburn, AL, Aug. 2025.
- **K. Thompson**, B. Jones, S. Martin, and D. Bevly, "GPS-Independent Autonomous Vehicle Convoying with UWB Ranging and Vehicle Models," *NDIA Ground Vehicle Systems Engineering and Technology Symposium (GVSETS)*, Novi, MI, Aug. 2022.

Skills

Guidance and Control: Classical control, LQR, MPC, trajectory and raceline optimization, path tracking, Kalman filtering, embedded LQR and QP controllers
Languages: C++, Python, MATLAB; also C# (Windows), Swift (Apple), and Kotlin (Android)
Tools and Libraries: ROS/ROS 2, Qt, CasADi, IPOPT, OSQP, Git, and SolidWorks (3D CAD)
Computer and Software: Linux, Windows, macOS, LaTeX, Visual Studio, Xcode, Single Board Computers and Microcontrollers

Achievements and Involvements

- Dean's List – Fall 2015, Fall 2016, Spring 2017, Fall 2018, Spring 2020 (Auburn University)
- Eagle Scout – Boy Scouts of America, 2013 (Troop 8 – Madison, Alabama)